

Technical Supplement to

Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

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1 Reaction matrices, flux cone, and semipositive conservation

Order species as $X_1, \dots, X_m, Z$ and reactions as $R_0, R_2, \dots, R_{m-2}, R_a, R_b, R_+, R_-$. For each indexed reaction, the source and target vectors have total molecularity at most two. The source matrix Y_m and stoichiometric matrix Γ_m are obtained by placing these source vectors and reaction differences in columns.

The balance equations $\Gamma_m v = 0$ read, successively, equality of every flux around the long circuit and equality of the two boundary-pair fluxes. Thus

$$\ker \Gamma_m = \text{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

An $m \times m$ minor formed by deleting one circuit reaction and one member of the reversible pair is triangular after elementary row operations and has absolute determinant 2^{m-2} . Therefore $\text{rank } \Gamma_m = m$. The vector

$$c = (0, 4, \dots, 4, 2, 1)^T$$

is orthogonal to every reaction vector, so $\text{im } \Gamma_m = c^\perp$. Notice that c is semipositive rather than positive: X_1 is absent.

For flux $(a\mathbf{1}_m, b, b)$ and unit equilibrium, $A_m(a, b) = \Gamma_m \text{diag}(a\mathbf{1}_m, b, b)Y_m^T$ has precisely the entries printed in the main text. At an arbitrary positive equilibrium, right multiplication by $H = \text{diag}(1/x_i^*)$ gives $J = A_m(a, b)H$. Conversely the rate assignment $k_r = v_r/(x^*)^{y_r}$ realizes every $a, b, H > 0$.

2 All-spectrum principal-block classification

The induced directed graph has the forward chain $X_2 \rightarrow X_3 \rightarrow \cdots \rightarrow X_{m-1}$, boundary arrows involving X_1, X_m, Z , and no alternative route that can close an interior cycle. A nontrivial strongly connected induced block with fewer than m vertices is therefore one of the two complete long cycles, a boundary-triad block, or a negative singleton. At $b = 2a$, the coefficient A_{m1} vanishes; deleting that edge can only split a component and cannot create a new component.

For the boundary triad, the Routh–Hurwitz difference expands as

$$\begin{aligned} c_1 c_2 - c_3 = & 4a^3 h_1^2 h_m + 16a^3 h_1 h_m^2 + 4a^2 b h_1^2 h_m + 23a^2 b h_1 h_m^2 \\ & + 16a^2 b h_1 h_m h_Z + 4a^2 b h_1^2 h_Z + 64a^2 b h_m^2 h_Z \\ & + 7ab^2 h_1^2 h_m + 7ab^2 h_1 h_m^2 + 28ab^2 h_1 h_m h_Z \\ & + 4ab^2 h_1^2 h_Z + 20ab^2 h_1 h_m h_Z + 16ab^2 h_m^2 h_Z + 64ab^2 h_m h_Z^2. \end{aligned}$$

Every monomial is strictly positive. The two long-cycle modulus inequalities are strict because $a + b > a$ and $4a + b > 4a$. This proves Hurwitz stability of every smaller principal block.

Sparse expansion on $\{X_1, \dots, X_m\}$ gives

$$(-1)^m \det J_{\widehat{Z}, \widehat{Z}} = -2a^{m-1} b \prod_{i=1}^m h_i.$$

The complete omission table is derived in Section S4 below.

3 The one-bad-minor theorem in full detail

Let $a_I = (-1)^{|I|} \det J_{I, I}$. Multilinearity in the diagonal entries of sD gives

$$\det(sD - J) = \sum_{I \subseteq [n]} a_I \prod_{j \notin I} s d_j.$$

Regrouping by $k = n - |I|$ yields

$$\det(sD - J) = \sum_{k=1}^n \beta_k(D) s^k, \quad \beta_k(D) = \sum_{|I|=n-k} a_I \prod_{j \notin I} d_j.$$

For $k \geq 2$, every contributing principal block has order at most $n - 2$ and is Hurwitz, so $a_I > 0$. Hence $\beta_k > 0$.

The bracket $q_D(s) = \beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}$ is strictly increasing for $s > 0$. If $\beta_1 < 0$, it has exactly one positive zero and its derivative there is positive. Since $\det(sD - J) = \det(D) \det(sI - D^{-1}J)$, the zero is algebraically simple.

For the positive-real-eigenvalue band, differentiate

$$\chi_s(\lambda) = \sum_I a_I \prod_{j \notin I} (\lambda + s d_j).$$

Every order-at-most- $(n - 2)$ contribution is nonnegative for $\lambda \geq 0$. The order- $(n - 1)$ contribution to χ'_s is the constant $\sum_{|I|=n-1} a_I$. This equals the coefficient of s in $\det(sI - J)$ and is positive because equal damping sends the simple zero to $-s$ and shifts each stable eigenvalue by $-s$. Thus $\chi'_s > 0$ on the nonnegative real axis. The sign of $\chi_s(0)$ then determines whether there is exactly one positive real eigenvalue.

The theorem says nothing about nonreal unstable eigenvalues when $s > s_*$. The all-mode certificates below are separate, profile-specific results.

4 Order- $(n - 1)$ omission minors and diffusion design

For $J = A_m(a, b)H$, omitting Z gives

$$(-1)^m \det J_{\widehat{Z}, \widehat{Z}} = -2a^{m-1} b \prod_{i=1}^m h_i.$$

For $2 \leq j \leq m-1$, omitting X_j breaks the chain. Each resulting one-way component contributes a negative diagonal factor, while the surviving boundary component has signed determinant $16a^2b$. Therefore

$$(-1)^m \det J_{\widehat{X_j}, \widehat{X_j}} = 16a^{m-1}bh_Z \prod_{i \neq j} h_i.$$

Omitting X_1 leaves the restricted conservation vector in the left kernel. Omitting X_m leaves the restriction of $H^{-1}(2, -2, \dots, -2, 0, 1)^T$ in the right kernel. Hence both determinants vanish.

Multiplication by the omitted diffusivities produces

$$\beta_1(D) = 2a^{m-1}b \left(\prod_{i=1}^m h_i \right) \left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z \right).$$

This yields the exact diffusion criterion and explains the coefficient eight as $16/2$.

For fixed H , if $d_{\min} = \min_i d_i$, then

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} \geq d_{\min} T(H),$$

so $\chi_D > T(H)$. Setting all X diffusivities to one and $d_Z = T(H) + \varepsilon$ proves sharpness as a nonattained infimum. Since $h_Z/h_j \geq 1/\chi_H$, one obtains $\chi_D \chi_H > 8(m-2)$.

5 Improved unit-equilibrium critical profile

Put $K_i = 91m - 181 - i$. The right vector and diffusion profile are

$$r_1 = 1, \quad r_i = -\frac{K_i}{63(m-2)}, \quad r_m = -\frac{2}{9}, \quad r_Z = \frac{5}{14},$$

$$d_1 = \frac{23}{63}, \quad d_i = \frac{1}{K_i}, \quad d_m = \frac{1}{7}, \quad d_Z = \frac{16}{45}.$$

Direct substitution gives $(A_m - D_m)r = 0$. With $\ell_Z = 1$, the left vector is

$$\ell_1 = -\frac{266}{815}, \quad \ell_i = \frac{78260(m-2)}{163(91m-180-i)}, \quad \ell_m = \frac{18368}{7335}, \quad \ell_Z = 1.$$

Define

$$H_m = \sum_{j=1}^{m-2} \frac{1}{91m-181-j}.$$

Then

$$\ell^T r = -\frac{7043400m - 13600927 - 7043400H_m}{924210} < 0,$$

$$\ell^T D_m r = -\frac{2(1760850H_m + 16559)}{462105} < 0.$$

The harmonic bound $H_m \leq (m-2)/(90m-179)$ makes the first sign an elementary shifted-polynomial inequality.

Homogeneous and spatial determinants

The homogeneous determinant has

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \quad \lambda q_m = (1 + \lambda)^{m-3} P(\lambda) - R(\lambda),$$

with $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. For a mode with damping factor t ,

$$\det(\lambda I - A_m + tD_m) = Q_m F - G,$$

where

$$Q_m = \prod_{i=2}^{m-1} \left(\lambda + 1 + \frac{t}{K_i} \right),$$

$$g_1 = \lambda + 2 + \frac{23}{63}t, \quad g_m = \lambda + 5 + \frac{1}{7}t, \quad g_Z = \lambda + 4 + \frac{16}{45}t,$$

$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \quad G = g_Z(4g_1 + g_m) - 36.$$

At onset, $\prod_{i=2}^{m-1} (1 + K_i^{-1}) = 91/90$. The exact certificate tables appear in Section S8.

6 Conservation-compatible stable-mode corrections

The quadratic tensor is obtained reaction by reaction. Its components are

$$\begin{aligned} (B(u, v))_1 &= -(u_1 v_{m-1} + u_{m-1} v_1) + 2u_Z v_Z - (u_1 v_m + u_m v_1), \\ (B(u, v))_2 &= -(u_1 v_2 + u_2 v_1) + 2u_m v_m, \\ (B(u, v))_i &= u_1 v_{i-1} + u_{i-1} v_1 - u_1 v_i - u_i v_1, \\ (B(u, v))_m &= 2(u_1 v_{m-1} + u_{m-1} v_1) - 4u_m v_m + 2u_Z v_Z - (u_1 v_m + u_m v_1), \\ (B(u, v))_Z &= -4u_Z v_Z + 2(u_1 v_m + u_m v_1). \end{aligned}$$

Here the fourth line holds for $3 \leq i \leq m-1$. Reaction-wise conservation implies $c^T B(u, v) = 0$.

The zero mode solves

$$A_m w_0 = -\frac{1}{4}B(r, r), \quad c^T w_0 = 0.$$

The exact solution is

$$\begin{aligned} (w_0)_1 &= \frac{182448m - 373417}{31752(8m - 17)}, \\ (w_0)_2 &= \frac{1008m^2 - 20459m + 37138}{31752(m - 2)(8m - 17)}, \\ (w_0)_i &= (w_0)_2 - \frac{i - 2}{126(m - 2)}, \quad (w_0)_m = -\frac{1}{81}, \quad (w_0)_Z = \frac{16861m - 34044}{7938(8m - 17)}. \end{aligned}$$

The second harmonic solves

$$(A_m - 4D_m)w_2 = -\frac{1}{4}B(r, r).$$

With

$$\sigma = \frac{1}{126(m - 2)}, \quad T_i = \frac{K_{i-3}K_{i-2}K_{i-1}K_i}{K_{-1}K_0K_1K_2},$$

the chain recurrence is

$$(w_2)_i = T_i \left((w_2)_2 + \frac{\sigma K_2}{3} \right) - \frac{\sigma K_i}{3}.$$

Its boundary denominator is

$$Q_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500.$$

After $m = u + 3$, the coefficients are

$$589180301, \quad 1802606769, \quad 1838302066, \quad 624878904,$$

so it never vanishes for $m \geq 3$. The complete boundary values are provided in the machine-readable certificate and in the public verifier.

The cubic coefficient is

$$c_m = \frac{\ell^T \{B(r, w_0) + \frac{1}{2}B(r, w_2)\}}{\ell^T r}.$$

The numerator equals $R_m + C_m H_m$, where $R_m > 0$ and $C_m < 0$. Substitution of $H_m \leq (m - 2)/(90m - 179)$ reduces positivity to the degree-five polynomial

$$L_m = 2729945147827667886720m^5 - 27755132420474170999952m^4$$

$$\begin{aligned}
& + 112813395868533457497683m^3 - 229153280695458887386228m^2 \\
& + 232620996871721820873517m - 94412163900120968220300.
\end{aligned}$$

After $m = u + 3$, its coefficients are

$$\begin{aligned}
& 2729945147827667886720, 13194044796940847300848, 25446870127333515303059, \\
& 24475321329207325509911, 11736484608366875120374, 2243933770033305838488,
\end{aligned}$$

all positive. Hence the cubic numerator is positive and, since $\ell^T r < 0$, $c_m < 0$.

7 Equilibrium-scaled stable family

Let $r = m - 2$, $L \in [1/\sqrt{3r}, 90r/(90r + 1)]$, and

$$h_1 = h_m = h_Z = 1, \quad h_i = \frac{K_i}{LK_{i-1}}.$$

The physical equilibrium is $x^* = H^{-1}\mathbf{1}$ and the physical diffusion matrix is $D^{\text{phys}} = H\Delta$, where Δ is the improved unit profile. In normalized variables $z = Hx$,

$$z_t = H\{f_m(z) + \Delta z_{\xi\xi}\}.$$

The stationary bracket and critical right vector are unchanged. The dynamic left vector is $H^{-1}\ell$ and the physical conservation gauge uses $H^{-1}c$.

For interior chain factors,

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

When $t \geq 1$ and $\Re\lambda \geq 0$,

$$|Q_m|^2 \geq \left(\frac{91}{90}\right)^2 |1 + L\lambda|^{2r} \geq \left(\frac{91}{90}\right)^2 \left(1 + 2rL\Re\lambda + \frac{1}{3}(\Im\lambda)^2\right).$$

The 34- and 84-term coefficient tables in Section S8 prove homogeneous and all-mode stability.

The physical zero-mode gauge is

$$w_0(L) = w_0^{\text{ref}} + \tau_m(L)\rho, \quad \tau_m(L) = -\frac{(H^{-1}c)^T w_0^{\text{ref}}}{(H^{-1}c)^T \rho}.$$

The second harmonic is unchanged. The cubic numerator satisfies

$$\begin{aligned}
N_m(L) &= N_m^{\text{ref}} + \tau_m(L)S_m, \\
N_m^{\text{ref}} &> \frac{1}{100}, \quad -\frac{1}{10} < S_m < 0, \quad \tau_m(L) < \frac{1}{20},
\end{aligned}$$

so $N_m(L) > 1/200$. Also

$$\ell^T H^{-1}r = -\frac{485873}{924210} - \frac{11180}{1467}L(m-2) < 0.$$

Therefore the family is uniformly supercritical over the certified interval.

The contrasts are

$$\chi_D(L) = \frac{23}{63}(91rL), \quad \chi_H(L) = \frac{91r-1}{91rL},$$

with constant product $23(91r-1)/63$. At $L = 1/\sqrt{3r}$, both have square-root growth.

8 Exact coefficient tables

Each row below records the exponent vector and exact coefficient of the corresponding polynomial. In the Pareto tables, the coefficient column is a polynomial in $A = 2rL\Re\lambda$ (or any positive lower coefficient replacing it). All listed coefficients are nonnegative; the strict first-order rows identify the unique equality case. The command `python verify_symbolic_certificates.py` regenerates and checks the tables from exact symbolic expressions.

35-term homogeneous certificate

$\deg_x$	$\deg_z$	coefficient
10	0	1
9	0	26
8	1	5
8	0	277
7	1	104
7	0	1582
6	2	10
6	1	892
6	0	5356
5	2	156
5	1	4034
5	0	11282
4	3	10
4	2	1014
4	1	10432
4	0	15116
3	3	104
3	2	3322
3	1	15628
3	0	12612
2	4	5
2	3	460
2	2	5804
2	1	13296
2	0	5568
1	4	26
1	3	870
1	2	4858
1	1	5700
1	0	960
0	5	1
0	4	61
0	3	728
0	2	1508
0	1	192

77-term improved-profile spatial certificate

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
6	1	0	1
6	0	0	<u>8281</u>
5	1	1	<u>8100</u>
5	1	0	<u>544</u>
5	0	1	<u>315</u>
5	0	0	<u>7474</u>
5	0	0	<u>315</u>
5	0	0	<u>160888</u>
5	0	0	<u>91125</u>
5	0	0	<u>4420871</u>
4	2	0	<u>182250</u>
4	1	2	3
4	1	2	<u>40058</u>
4	1	1	<u>33075</u>
4	1	1	<u>1147126</u>
4	1	0	<u>33075</u>
4	1	0	<u>9681647</u>
4	0	2	<u>44100</u>
4	0	2	<u>3384901</u>
4	0	1	<u>2733750</u>
4	0	1	<u>96932147</u>
4	0	0	<u>2733750</u>
4	0	0	<u>1073317517</u>
			<u>5467500</u>

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
3	2	1	<u>1088</u>
3	2	0	<u>315</u> <u>14948</u>
3	1	3	<u>315</u> <u>2744576</u>
3	1	2	<u>6251175</u> <u>41276762</u>
3	1	1	<u>2083725</u> <u>8186375738</u>
3	1	0	<u>31255875</u> <u>10434543841</u>
3	0	3	<u>10418625</u> <u>115958336</u>
3	0	2	<u>258339375</u> <u>3487886389</u>
3	0	1	<u>172226250</u> <u>39671949511</u>
3	0	0	<u>172226250</u> <u>315432259813</u>
2	3	0	<u>516678750</u> <u>3</u>
2	2	2	<u>147968</u> <u>99225</u>
2	2	1	<u>4065856</u> <u>99225</u>
2	2	0	<u>112938659</u> <u>396900</u>
2	1	4	<u>33947257</u> <u>393824025</u>
2	1	3	<u>2164641406</u> <u>393824025</u>
2	1	2	<u>1130666508023</u> <u>9845600625</u>
2	1	1	<u>9041228678666</u> <u>9845600625</u>
2	1	0	<u>5074377793279</u> <u>2187911250</u>
2	0	4	<u>5737086433</u> <u>65101522500</u>
2	0	3	<u>182912198807</u> <u>32550761250</u>
2	0	2	<u>2142193804481</u> <u>21700507500</u>
2	0	1	<u>17194597630733</u> <u>32550761250</u>
2	0	0	<u>34399269232129</u> <u>65101522500</u>
1	3	1	<u>544</u> <u>315</u>
1	3	0	<u>7474</u> <u>315</u>
1	2	3	<u>1817216</u> <u>6251175</u>
1	2	2	<u>24359162</u> <u>2083725</u>
1	2	1	<u>5056135954</u> <u>31255875</u>
1	2	0	<u>47437544293</u> <u>62511750</u>
1	1	5	<u>3399584</u> <u>393824025</u>
1	1	4	<u>57943048</u> <u>78764805</u>
1	1	3	<u>1934522477168</u> <u>88610405625</u>
1	1	2	<u>16308922895827</u> <u>59073603750</u>
1	1	1	<u>85574440190593</u> <u>59073603750</u>
1	1	0	<u>62146993882837</u> <u>25317258750</u>
1	0	5	<u>143632424</u> <u>16275380625</u>
1	0	4	<u>2448093778</u> <u>3255076125</u>
1	0	3	<u>98527787884</u> <u>5425126875</u>
1	0	2	<u>475980699652</u> <u>3255076125</u>
1	0	1	<u>4488365319374</u> <u>16275380625</u>
1	0	0	<u>1086412028</u> <u>10333575</u>
0	4	0	<u>1</u> <u>27794</u>
0	3	2	<u>99225</u> <u>624478</u>
0	3	1	<u>99225</u> <u>1747307</u>
0	3	0	<u>26460</u> <u>1744613</u>
0	2	4	<u>78764805</u> <u>437610814</u>
0	2	3	<u>393824025</u> <u>452220491743</u>
0	2	2	<u>19691201250</u> <u>4035916860091</u>
0	2	1	<u>19691201250</u> <u>10861695961357</u>
0	2	0	<u>13127467500</u> <u>135424</u>
0	1	6	<u>393824025</u> <u>2964608</u>
0	1	5	<u>78764805</u> <u>962538972893</u>
0	1	4	<u>637994920500</u> <u>43240103983367</u>
0	1	3	<u>1594987301250</u> <u>238300444324673</u>
0	1	2	<u>1063324867500</u>

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
0	1	1	$\frac{170939874507539}{227855328750}$
0	1	0	$\frac{32638690751329}{65101522500}$
0	0	6	$\frac{5721664}{16275380625}$
0	0	5	$\frac{125254688}{3255076125}$
0	0	4	$\frac{6479423564}{5425126875}$
0	0	3	$\frac{41256502232}{3255076125}$
0	0	2	$\frac{513831327844}{16275380625}$
0	0	1	$\frac{148103696}{10333575}$

34-term Pareto homogeneous certificate

$\deg_x$	$\deg_z$	coefficient
9	0	A
8	1	$\frac{1}{3}$
8	0	$1 + 24A$
7	1	$8 + 4A$
7	0	$24 + 228A$
6	2	$\frac{4}{3}$
6	1	$80 + 72A$
6	0	$228 + 1102A$
5	2	$24 + 6A$
5	1	$\frac{1318}{3} + 516A$
5	0	$1102 + 2924A$
4	3	2
4	2	$178 + 72A$
4	1	$\frac{4472}{3} + 1828A$
4	0	$2899 + 4332A$
3	3	$24 + 4A$
3	2	$\frac{2044}{3} + 348A$
3	1	$3272 + 3336A$
3	0	$4002 + 3553A$
2	4	$\frac{4}{3}$
2	3	$120 + 24A$
2	2	$1460 + 726A$
2	1	$\frac{13411}{3} + 2796A$
2	0	$2304 + 1504A$
1	4	$8 + A$
1	3	$266 + 60A$
1	2	$1658 + 668A$
1	1	$\frac{8902}{3} + 865A$
1	0	$448 + 256A$
0	5	$\frac{1}{3}$
0	4	21
0	3	$\frac{848}{3}$
0	2	$\frac{2794}{3}$
0	1	$\frac{64}{3}$

84-term Pareto spatial certificate

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
7	0	0	$\frac{8281}{8100}A$
6	1	0	$\frac{8281}{24300}$
6	0	1	$\frac{160888}{91125}A$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
6	0	0	$\frac{8281}{8100} + \frac{4420871}{182250} A$
5	1	1	$\frac{160888}{273375} A$
5	1	0	$\frac{4420871}{546750} + \frac{8281}{2700} A$
5	0	2	$\frac{3384901}{2733750} A$
5	0	1	$\frac{160888}{91125} + \frac{96932147}{2733750} A$
5	0	0	$\frac{4420871}{182250} + \frac{1210005017}{5467500} A$
4	2	0	$\frac{8281}{8100} A$
4	1	2	$\frac{3384901}{8201250} A$
4	1	1	$\frac{96932147}{8201250} + \frac{321776}{91125} A$
4	1	0	$\frac{315078023}{4100625} + \frac{4420871}{91125} A$
4	0	3	$\frac{115958336}{258339375} A$
4	0	2	$\frac{3384901}{2733750} + \frac{3487886389}{172226250} A$
4	0	1	$\frac{96932147}{2733750} + \frac{45494837011}{172226250} A$
4	0	0	$\frac{1073317517}{5467500} + \frac{503404909813}{516678750} A$
3	2	1	$\frac{321776}{273375} A$
3	2	0	$\frac{4420871}{273375} + \frac{8281}{2700} A$
3	1	3	$\frac{115958336}{775018125} A$
3	1	2	$\frac{3487886389}{516678750} + \frac{6251648}{4100625} A$
3	1	1	$\frac{47319306931}{516678750} + \frac{171782416}{4100625} A$
3	1	0	$\frac{578603925523}{1550036250} + \frac{2360113561}{8201250} A$
3	0	4	$\frac{5737086433}{65101522500} A$
3	0	3	$\frac{115958336}{258339375} + \frac{182912198807}{32550761250} A$
3	0	2	$\frac{3487886389}{172226250} + \frac{2513945531981}{21700507500} A$
3	0	1	$\frac{39671949511}{172226250} + \frac{29165273350733}{32550761250} A$
3	0	0	$\frac{315432259813}{516678750} + \frac{138537167092129}{65101522500} A$
2	3	0	$\frac{8281}{8100} A$
2	2	2	$\frac{6251648}{12301875} A$
2	2	1	$\frac{171782416}{12301875} + \frac{160888}{91125} A$
2	2	0	$\frac{4871148347}{49207500} + \frac{4420871}{182250} A$
2	1	4	$\frac{5737086433}{195304567500} A$
2	1	3	$\frac{182912198807}{97652283750} + \frac{76777376}{258339375} A$
2	1	2	$\frac{2613196695629}{65101522500} + \frac{2058349189}{172226250} A$
2	1	1	$\frac{33256099805357}{97652283750} + \frac{28172026051}{172226250} A$
2	1	0	$\frac{184975683058783}{195304567500} + \frac{388034017813}{516678750} A$
2	0	5	$\frac{143632424}{16275380625} A$
2	0	4	$\frac{5737086433}{65101522500} + \frac{2448093778}{3255076125} A$
2	0	3	$\frac{182912198807}{32550761250} + \frac{119438423884}{5425126875} A$
2	0	2	$\frac{2142193804481}{21700507500} + \frac{878963880652}{3255076125} A$
2	0	1	$\frac{17194597630733}{32550761250} + \frac{21944279796374}{16275380625} A$
2	0	0	$\frac{34399269232129}{65101522500} + \frac{21842305628}{10333575} A$
1	3	1	$\frac{160888}{273375} A$
1	3	0	$\frac{4420871}{546750} + \frac{8281}{8100} A$
1	2	3	$\frac{76777376}{775018125} A$
1	2	2	$\frac{2058349189}{516678750} + \frac{2348593}{8201250} A$
1	2	1	$\frac{29084261011}{516678750} + \frac{52768391}{8201250} A$
1	2	0	$\frac{212816762834}{775018125} + \frac{1090212071}{16402500} A$
1	1	5	$\frac{143632424}{48826141875} A$
1	1	4	$\frac{2448093778}{97652283750} + \frac{294839597}{13020304500} A$
1	1	3	$\frac{124375398572}{16275380625} + \frac{36978113783}{32550761250} A$
1	1	2	$\frac{9956792796683}{97652283750} + \frac{503148512477}{21700507500} A$
1	1	1	$\frac{11312104230233}{195304567500} + \frac{6606581721077}{32550761250} A$
1	1	0	$\frac{241757054979}{221433750} + \frac{52308962411329}{65101522500} A$
1	0	6	$\frac{5721664}{16275380625} A$
1	0	5	$\frac{143632424}{16275380625} + \frac{125254688}{3255076125} A$
1	0	4	$\frac{2448093778}{32550761250} + \frac{8242156364}{5425126875} A$
1	0	3	$\frac{98527787884}{5425126875} + \frac{86436565592}{3255076125} A$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
1	0	2	$\frac{475980699652}{3255076125} + \frac{3438519744244}{16275380625} A$
1	0	1	$\frac{4488365319374}{16275380625} + \frac{7094103536}{10333575} A$
1	0	0	$\frac{1086412028}{10333575} + \frac{4999696}{6561} A$
0	4	0	$\frac{8281}{24300}$
0	3	2	$\frac{2348593}{24603750}$
0	3	1	$\frac{52768391}{24603750}$
0	3	0	$\frac{570259573}{24603750}$
0	2	4	$\frac{294839597}{39060913500}$
0	2	3	$\frac{36978113783}{97652283750}$
0	2	2	$\frac{521791643711}{521791643711}$
0	2	1	$\frac{65101522500}{3617447476357}$
0	2	0	$\frac{48826141875}{30203751676613}$
0	1	6	$\frac{97652283750}{5721664}$
0	1	5	$\frac{48826141875}{125254688}$
0	1	4	$\frac{11480941147}{9765228375}$
0	1	3	$\frac{21700507500}{975299997269}$
0	1	2	$\frac{97652283750}{17163191001769}$
0	1	1	$\frac{195304567500}{10010796373877}$
0	1	0	$\frac{32550761250}{650029271329}$
0	0	6	$\frac{65101522500}{5721664}$
0	0	5	$\frac{16275380625}{125254688}$
0	0	4	$\frac{3255076125}{6479423564}$
0	0	3	$\frac{5425126875}{41256502232}$
0	0	2	$\frac{3255076125}{513831327844}$
0	0	1	$\frac{16275380625}{148103696}$
0	0	0	$\frac{10333575}{10333575}$

9 Near-threshold control example

For the affine path

$$p = \varepsilon, \quad u = 1 + (2 - t)\varepsilon + \theta\varepsilon^2, \quad v = t\varepsilon - \theta\varepsilon^2,$$

$$q = \frac{1}{2} - \left(\frac{1}{2} + t\right)\varepsilon + \left(\theta - \frac{M}{2}\right)\varepsilon^2,$$

the critical vector tends to one half of the homogeneous kernel vector. The diffusion excess over the exact linear threshold is

$$d_Z - 8 \sum_{j=2}^{m-1} d_j = 4 \left(M + 6t + 3t^2 - \frac{t^2}{m-2} \right) \varepsilon^2 + O(\varepsilon^3).$$

For $(t, \theta, M) = (2/9, 1/2, 1)$ and $m = 3$, exact elimination yields

$$c_3(\varepsilon) = \frac{6}{1379} + \frac{421985}{11409846} \varepsilon + O(\varepsilon^2),$$

and rational remainder bounds prove $c_3(\varepsilon) > 0$ on $0 < \varepsilon \leq 10^{-3}$. This is a rigorous subcritical control example, not a universal nonlinear lower bound.

10 Semilinear stability and robustness

On the real fixed-integrated-mass L^2 space, the positive diagonal Neumann diffusion operator has domain H_N^2 , is sectorial, and generates an analytic semigroup. Its half-order fractional domain is H^1 . In one space dimension, H^1 is a Banach algebra and embeds into C^0 , so the quadratic mass-action Nemytskii map is smooth from H^1 to L^2 . The patterned branch lies in H_N^2 . The fixed-mass restriction removes the homogeneous conservation zero, and Neumann boundary conditions on an interval leave no continuous translation-neutral mode. Strict spectral negativity therefore implies local exponential asymptotic stability in H^1 .

For robustness, use one scalar diffusion multiplier as the implicit variable. The critical eigenvalue derivative is nonzero, so the implicit function theorem continues the crossing under small perturbations of rates, equilibrium

coordinates, diffusion ratios, and interval length. Low-mode gaps persist by analytic perturbation. A positive diffusion lower bound and a reaction-Jacobian upper bound give one neighborhood controlling all high modes. Smooth dependence of the Lyapunov–Schmidt coefficients preserves the negative cubic sign and the branch spectral gap. This is a retuned codimension-one statement and is not a claim that every nearby point is itself at bifurcation.

11 Numerical protocol and independent verification

Cosine–Galerkin simulations use exact modal convolution for the quadratic reaction terms and an adaptive stiff integrator. Initial data are small perturbations of **1** in the critical direction. The first-mode amplitude is measured by the adjoint projection. Spatial and temporal refinement tables, open-format profile data, exact parameters, and all solver tolerances are included in the data archive. No numerical output is used in a proof.

The release contains independent programs for the reaction family, flux cone, all-spectrum block theorem, omission minors, one-bad-minor theorem, diffusion criterion, contrast bounds, improved profile, Pareto family, mode certificates, harmonic corrections, cubic sign, and branch stability. Mutation tests change signs, endpoints, the factor eight, Fourier factors, and certificate coefficients and must be rejected.